SRI: N04LO

The spectra are evaluated at $-k_*\eta_* = 1$, through relative amplitude order $p = 4$. The Hubble-flow parameters are denoted by ϵ_j and $M_{\rm Pl}$ is the reduced Planck mass. Coefficients are generated numerically from the normalized mode equations, using the iterative algebra. The convention for the spectral derivatives is

$$\alpha_n^A = \frac{d^{n+1}\ln P_A}{d(\ln k)^{n+1}}, \qquad \alpha_0^s = n_s - 1, \qquad \alpha_0^t = n_t, \qquad r = P_t/P_s.$$

1 Spectra

1.1 Scalar spectrum

$$P_s(k_*) = \frac{H_*^2}{8\pi^2 M_{\rm Pl}^2 \epsilon_{1*}} \left[\sum_{d=0}^{4} F_{s,d} + O(\mathrm{SR}^5)\right].$$

Relative order 0.

$$F_{s,0} = 1.0$$

Relative order 1.

$$F_{s,1} = 0.72963715453852\epsilon_2 - 0.54072569092296\epsilon_1$$

Relative order 2.

$$\begin{aligned} F_{s,2} = {} & 1.5402686460338\epsilon_1^2 + 0.49988573877771\epsilon_2^2 \\ & + 1.0192767657904\epsilon_1\epsilon_2 + 0.14504832807052\epsilon_2\epsilon_3 \end{aligned}$$

Relative order 3.

$$\begin{aligned} F_{s,3} = {} & 0.20072294544592\epsilon_2^3 + 1.6057835635673\epsilon_1^3 \\ & + 4.2977813647689\epsilon_1^2\epsilon_2 + 1.959600275274\epsilon_1\epsilon_2^2 \\ & + 0.030669313835717\epsilon_2\epsilon_3^2 - 0.23520055946694\epsilon_2^2\epsilon_3 \\ & + 0.030669313835717\epsilon_2\epsilon_3\epsilon_4 - 0.65969152604436\epsilon_1\epsilon_2\epsilon_3 \end{aligned}$$

Relative order 4.

$$\begin{aligned} F_{s,4} = {} & 0.083467005785127\epsilon_2^4 + 2.9412556561294\epsilon_1^4 \\ & + 0.055787239966231\epsilon_2\epsilon_3^3 + 8.1576664436908\epsilon_1^2\epsilon_2^2 \\ & + 9.1655563981061\epsilon_1^3\epsilon_2 + 2.2830486720555\epsilon_1\epsilon_2^3 \\ & - 0.20996470057763\epsilon_2^3\epsilon_3 - 0.058660421836305\epsilon_2^2\epsilon_3^2 \\ & + 0.055787239966231\epsilon_2\epsilon_3\epsilon_4^2 + 0.16736171989869\epsilon_2\epsilon_3^2\epsilon_4 \\ & + 0.38664546255166\epsilon_1^2\epsilon_2\epsilon_3 - 0.016062214045699\epsilon_2^2\epsilon_3\epsilon_4 \\ & - 0.079171041433698\epsilon_1\epsilon_2\epsilon_3^2 - 0.91389525644738\epsilon_1\epsilon_2^2\epsilon_3 \\ & + 0.055787239966231\epsilon_2\epsilon_3\epsilon_4\epsilon_5 \\ & - 0.079171041433698\epsilon_1\epsilon_2\epsilon_3\epsilon_4 \end{aligned}$$

1.2 Tensor spectrum

$$P_t(k_*) = \frac{2H_*^2}{\pi^2 M_{\rm Pl}^2} \left[\sum_{d=0}^{4} F_{t,d} + O(\mathrm{SR}^5)\right].$$

Relative order 0.

$$F_{t,0} = 1.0$$

Relative order 1.

$$F_{t,1} = -0.54072569092296\epsilon_1$$

Relative order 2.

$$F_{t,2} = 1.5402686460338\epsilon_1^2 - 0.25062903478191\epsilon_1\epsilon_2$$

Relative order 3.

$$\begin{aligned} F_{t,3} = {} & 1.6057835635673\epsilon_1^3 + 1.8891060194179\epsilon_1^2\epsilon_2 \\ & - 0.18929040711048\epsilon_1\epsilon_2^2 - 0.18929040711048\epsilon_1\epsilon_2\epsilon_3 \end{aligned}$$

Relative order 4.

$$\begin{aligned} F_{t,4} = {} & 2.9412556561294\epsilon_1^4 + 4.0859368676311\epsilon_1^3\epsilon_2 \\ & + 1.8087367742501\epsilon_1^2\epsilon_2^2 - 0.077715927178017\epsilon_1\epsilon_2^3 \\ & + 1.696905383796\epsilon_1^2\epsilon_2\epsilon_3 - 0.077715927178017\epsilon_1\epsilon_2\epsilon_3^2 \\ & - 0.23314778153405\epsilon_1\epsilon_2^2\epsilon_3 - 0.077715927178017\epsilon_1\epsilon_2\epsilon_3\epsilon_4 \end{aligned}$$

2 Consistency relations

2.1 $n = 0$

For $p = 4$, the absolute cutoff is $W = 5$:

$$\alpha_0^t = \sum_{d=1}^{5} R_{0,d} + O(\mathrm{SR}^6).$$

Absolute order 1.

$$R_{0,1} = -0.125r$$

Absolute order 2.

$$R_{0,2} = 0.0078125r^2 + 0.125\alpha_0^s r$$

Absolute order 3.

$$\begin{aligned} R_{0,3} = {} & -0.0030712246746565r^3 + 0.125\alpha_1^s r \\ & - 0.10280837917801\left(\alpha_0^s\right)^2 r - 0.041327094794504\alpha_0^s r^2 \end{aligned}$$

Absolute order 4.

$$\begin{aligned} R_{0,4} = {} & 0.0014926374043055r^4 + 0.125\alpha_2^s r \\ & + 0.065244100954277\left(\alpha_0^s\right)^3 r + 0.024554893470954\alpha_0^s r^3 \\ & + 0.10320649151137\left(\alpha_0^s\right)^2 r^2 - 0.060109233906372\alpha_1^s r^2 \\ & - 0.23087387125098\alpha_0^s\alpha_1^s r \end{aligned}$$

Absolute order 5.

$$\begin{aligned} R_{0,5} = {} & -0.0009328010098341 5r^5 + 0.125\alpha_3^s r \\ & + 0.04315951309091\alpha_1^s r^3 - 0.038536574146577\left(\alpha_0^s\right)^4 r \\ & - 0.018206214478227\alpha_0^s r^4 - 0.077020534433109\alpha_2^s r^2 \\ & - 0.14103704832205\left(\alpha_1^s\right)^2 r - 0.18695785481671\left(\alpha_0^s\right)^3 r^2 \\ & - 0.10777687087114\left(\alpha_0^s\right)^2 r^3 + 0.33322659757276\alpha_0^s\alpha_1^s r^2 \\ & + 0.25691700713838\left(\alpha_0^s\right)^2\alpha_1^s r - 0.24116427546487\alpha_0^s\alpha_2^s r \end{aligned}$$

2.2 $n = 1$

For $p = 4$, the absolute cutoff is $W = 6$:

$$\alpha_1^t = \sum_{d=2}^{6} R_{1,d} + O(\mathrm{SR}^7).$$

Absolute order 2.

$$R_{1,2} = 0.015625r^2 + 0.125\alpha_0^s r$$

Absolute order 3.

$$\begin{aligned} R_{1,3} = &-0.0029296875r^3 + 0.125\alpha_1^s r \\ &- 0.125\left(\alpha_0^s\right)^2 r - 0.046875\alpha_0^s r^2 \end{aligned}$$

Absolute order 4.

$$\begin{aligned} R_{1,4} = &\, 0.0016576826498282r^4 + 0.125\alpha_2^s r \\ &+ 0.12398128438351\left(\alpha_0^s\right)^2 r^2 + 0.10280837917801\left(\alpha_0^s\right)^3 r \\ &+ 0.027641022071908\alpha_0^s r^3 - 0.072577094794504\alpha_1^s r^2 \\ &- 0.33061675835603\alpha_0^s\alpha_1^s r \end{aligned}$$

Absolute order 5.

$$\begin{aligned} R_{1,5} = &-0.0010528680915447r^5 + 0.125\alpha_3^s r \\ &+ 0.050025543685844\alpha_1^s r^3 - 0.091359233906372\alpha_2^s r^2 \\ &- 0.065244100954277\left(\alpha_0^s\right)^4 r - 0.13027434666455\left(\alpha_0^s\right)^2 r^3 \\ &- 0.021075080402355\alpha_0^s r^4 - 0.23087387125098\left(\alpha_1^s\right)^2 r \\ &- 0.24842610305581\left(\alpha_0^s\right)^3 r^2 + 0.42660617411381\left(\alpha_0^s\right)^2\alpha_1^s r \\ &+ 0.41559991864823\alpha_0^s\alpha_1^s r^2 - 0.35587387125098\alpha_0^s\alpha_2^s r \end{aligned}$$

Absolute order 6.

$$\begin{aligned} R_{1,6} = &\, 0.00079786539870909r^6 + 0.125\alpha_4^s r \\ &+ 0.038536574146577\left(\alpha_0^s\right)^5 r + 0.41043044123784\left(\alpha_0^s\right)^4 r^2 \\ &+ 0.14689479099556\left(\alpha_0^s\right)^2 r^4 + 0.018569586968286\alpha_0^s r^5 \\ &+ 0.074971901003326\alpha_2^s r^3 + 0.38411085965327\left(\alpha_1^s\right)^2 r^2 \\ &+ 0.44641771599844\left(\alpha_0^s\right)^3 r^3 - 0.043199996920584\alpha_1^s r^4 \\ &- 0.10827053443311\alpha_3^s r^2 + 0.5788087353052\alpha_0^s\alpha_2^s r^2 \\ &+ 0.65487106259881\alpha_0^s\left(\alpha_1^s\right)^2 r + 0.49808128260325\left(\alpha_0^s\right)^2\alpha_2^s r \\ &- 0.36616427546487\alpha_0^s\alpha_3^s r - 0.52323837210896\alpha_1^s\alpha_2^s r \\ &- 0.51344198472507\alpha_0^s\alpha_1^s r^3 - 1.3749765739875\left(\alpha_0^s\right)^2\alpha_1^s r^2 \\ &- 0.41106330372469\left(\alpha_0^s\right)^3\alpha_1^s r \end{aligned}$$

2.3 $n = 2$

For $p = 4$, the absolute cutoff is $W = 7$:

$$\alpha_2^t = \sum_{d=3}^{7} R_{2,d} + O(\mathrm{SR}^8).$$

Absolute order 3.

$$\begin{aligned} R_{2,3} = & -0.00390625r^3 + 0.125\alpha_1^s r \\ & - 0.125\left(\alpha_0^s\right)^2 r - 0.046875\alpha_0^s r^2 \end{aligned}$$

Absolute order 4.

$$\begin{aligned} R_{2,4} = & \, 0.0013427734375r^4 + 0.125\alpha_2^s r \\ & + 0.125\left(\alpha_0^s\right)^3 r + 0.125\left(\alpha_0^s\right)^2 r^2 \\ & + 0.025390625\alpha_0^s r^3 - 0.0625\alpha_1^s r^2 \\ & - 0.375\alpha_0^s \alpha_1^s r \end{aligned}$$

Absolute order 5.

$$\begin{aligned} R_{2,5} = & -0.00099348164677838r^5 + 0.125\alpha_3^s r \\ & + 0.050668108270534\alpha_1^s r^3 - 0.13499234851023\left(\alpha_0^s\right)^2 r^3 \\ & - 0.28928966356152\left(\alpha_0^s\right)^3 r^2 - 0.020502543360439\alpha_0^s r^4 \\ & - 0.10280837917801\left(\alpha_0^s\right)^4 r - 0.33061675835603\left(\alpha_1^s\right)^2 r \\ & - 0.088202094794504\alpha_2^s r^2 + 0.63904189589007\left(\alpha_0^s\right)^2 \alpha_1^s r \\ & + 0.46569385315053\alpha_0^s \alpha_1^s r^2 - 0.45561675835603\alpha_0^s \alpha_2^s r \end{aligned}$$

Absolute order 6.

$$\begin{aligned} R_{2,6} = & \, 0.0007834832445242r^6 + 0.125\alpha_4^s r \\ & + 0.065244100954277\left(\alpha_0^s\right)^5 r + 0.51447193930681\left(\alpha_0^s\right)^3 r^3 \\ & + 0.018883632267472\alpha_0^s r^5 + 0.077748164662437\alpha_2^s r^3 \\ & + 0.5388653261447\left(\alpha_0^s\right)^4 r^2 + 0.15691226448912\left(\alpha_0^s\right)^2 r^4 \\ & + 0.4600841525546\left(\alpha_1^s\right)^2 r^2 - 0.10698423390637\alpha_3^s r^2 \\ & - 0.044329625847117\alpha_1^s r^4 + 0.78248004536479\left(\alpha_0^s\right)^2 \alpha_2^s r \\ & + 1.0840862194786\alpha_0^s \left(\alpha_1^s\right)^2 r + 0.67405262036735\alpha_0^s \alpha_2^s r^2 \\ & - 0.56686562073618\alpha_0^s \alpha_1^s r^3 - 0.68758257793092\left(\alpha_0^s\right)^3 \alpha_1^s r \\ & - 0.81762161375293\alpha_1^s \alpha_2^s r - 1.7284662943263\left(\alpha_0^s\right)^2 \alpha_1^s r^2 \\ & - 0.48087387125098\alpha_0^s \alpha_3^s r \end{aligned}$$

Absolute order 7.

$$\begin{aligned} R_{2,7} = & -0.00070216008727898r^7 + 0.125\alpha_5^s r \\ & + 0.10692234711116\alpha_3^s r^3 + 0.65487106259881\left(\alpha_1^s\right)^3 r \\ & + 0.044680319835584\alpha_1^s r^5 - 0.038536574146577\left(\alpha_0^s\right)^6 r \\ & - 0.076631375424485\alpha_2^s r^4 - 0.85737561408011\left(\alpha_0^s\right)^5 r^2 \\ & - 0.52323837210896\left(\alpha_2^s\right)^2 r - 0.019279820845041\alpha_0^s r^6 \\ & - 1.5778098507287\left(\alpha_0^s\right)^4 r^3 - 0.12389553443311\alpha_4^s r^2 \\ & - 0.64133873745452\left(\alpha_1^s\right)^2 r^3 - 0.193795881651\left(\alpha_0^s\right)^2 r^5 \\ & - 0.85944179815707\left(\alpha_0^s\right)^3 r^4 + 1.4436852511254\alpha_1^s \alpha_2^s r^2 \\ & + 4.6123541905271\left(\alpha_0^s\right)^3 \alpha_1^s r^2 + 0.72830526168963\alpha_0^s \alpha_1^s r^4 \\ & + 0.87237033860452\alpha_0^s \alpha_3^s r^2 + 0.60374617445758\left(\alpha_0^s\right)^4 \alpha_1^s r \\ & + 0.86424555806812\left(\alpha_0^s\right)^2 \alpha_3^s r + 3.481878776513\left(\alpha_0^s\right)^2 \alpha_1^s r^3 \end{aligned}$$

$$
\begin{aligned}
&- 0.88940264757383\alpha_1^s\alpha_3^s r - 0.94272853191881\alpha_0^s\alpha_2^s r^3 \\
&- 1.8880609737729\left(\alpha_0^s\right)^2\left(\alpha_1^s\right)^2 r - 0.49116427546487\alpha_0^s\alpha_4^s r \\
&- 2.69796002066\left(\alpha_0^s\right)^2\alpha_2^s r^2 - 0.90914458632794\left(\alpha_0^s\right)^3\alpha_2^s r \\
&- 3.7167089437539\alpha_0^s\left(\alpha_1^s\right)^2 r^2 + 2.8291430625131\alpha_0^s\alpha_1^s\alpha_2^s r
\end{aligned}
$$

2.4 $n = 3$

For $p = 4$, the absolute cutoff is $W = 8$:

$$
\alpha_3^t = \sum_{d=4}^{8} R_{3,d} + O(\mathrm{SR}^9).
$$

Absolute order 4.

$$
\begin{aligned}
R_{3,4} = {} & 0.00146484375r^4 + 0.125\alpha_2^s r \\
& + 0.125\left(\alpha_0^s\right)^3 r + 0.0234375\alpha_0^s r^3 \\
& + 0.109375\left(\alpha_0^s\right)^2 r^2 - 0.0625\alpha_1^s r^2 \\
& - 0.375\alpha_0^s\alpha_1^s r
\end{aligned}
$$

Absolute order 5.

$$
\begin{aligned}
R_{3,5} = {} & -0.000762939453125r^5 + 0.125\alpha_3^s r \\
& + 0.0419921875\alpha_1^s r^3 - 0.125\left(\alpha_0^s\right)^4 r \\
& - 0.375\left(\alpha_1^s\right)^2 r - 0.078125\alpha_2^s r^2 \\
& - 0.28125\left(\alpha_0^s\right)^3 r^2 - 0.01708984375\alpha_0^s r^4 \\
& - 0.1201171875\left(\alpha_0^s\right)^2 r^3 + 0.75\left(\alpha_0^s\right)^2\alpha_1^s r \\
& + 0.4375\alpha_0^s\alpha_1^s r^2 - 0.5\alpha_0^s\alpha_2^s r
\end{aligned}
$$

Absolute order 6.

$$
\begin{aligned}
R_{3,6} = {} & 0.00069887861331449r^6 + 0.125\alpha_4^s r \\
& + 0.52264594794504\left(\alpha_1^s\right)^2 r^2 + 0.10280837917801\left(\alpha_0^s\right)^5 r \\
& + 0.52433019631832\left(\alpha_0^s\right)^3 r^3 + 0.017257388611671\alpha_0^s r^5 \\
& + 0.07369519446916\alpha_2^s r^3 + 0.1495700174229\left(\alpha_0^s\right)^2 r^4 \\
& + 0.61990642191755\left(\alpha_0^s\right)^4 r^2 - 0.042328393296221\alpha_1^s r^4 \\
& - 0.1038270947945\alpha_3^s r^2 + 1.6087005501362\alpha_0^s\left(\alpha_1^s\right)^2 r \\
& + 1.0946586542461\left(\alpha_0^s\right)^2\alpha_2^s r + 0.71467513753404\alpha_0^s\alpha_2^s r^2 \\
& - 0.57385180946901\alpha_0^s\alpha_1^s r^3 - 1.9544879813691\left(\alpha_0^s\right)^2\alpha_1^s r^2 \\
& - 1.1168502750681\alpha_1^s\alpha_2^s r - 1.0502754126021\left(\alpha_0^s\right)^3\alpha_1^s r \\
& - 0.58061675835603\alpha_0^s\alpha_3^s r
\end{aligned}
$$

Absolute order 7.

$$
\begin{aligned}
R_{3,7} = {} & -0.00066040799045703r^7 + 0.125\alpha_5^s r \\
& + 0.044182253542082\alpha_1^s r^5 + 0.10547078563903\alpha_3^s r^3 \\
& + 1.0840862194786\left(\alpha_1^s\right)^3 r - 0.12260923390637\alpha_4^s r^2 \\
& - 0.065244100954277\left(\alpha_0^s\right)^6 r - 1.8011431833565\left(\alpha_0^s\right)^4 r^3 \\
& - 1.1197437723225\left(\alpha_0^s\right)^5 r^2 - 0.076712092161027\alpha_2^s r^4 \\
& - 0.70760825653778\left(\alpha_1^s\right)^2 r^3 - 0.19445546235151\left(\alpha_0^s\right)^2 r^5
\end{aligned}
$$

$$
\begin{aligned}
&- 0.018648400200653\alpha_0^s r^6 - 0.81762161375293\left(\alpha_2^s\right)^2 r \\
&- 0.9077746502428\left(\alpha_0^s\right)^3 r^4 + 0.75724388321833\alpha_0^s\alpha_1^s r^4 \\
&+ 1.2633539166158\left(\alpha_0^s\right)^2\alpha_3^s r + 5.8694148411763\left(\alpha_0^s\right)^3\alpha_1^s r^2 \\
&+ 3.899202830862\left(\alpha_0^s\right)^2\alpha_1^s r^3 + 1.0138030827023\left(\alpha_0^s\right)^4\alpha_1^s r \\
&+ 1.7276736271957\alpha_1^s\alpha_2^s r^2 + 0.96375532208646\alpha_0^s\alpha_3^s r^2 \\
&- 1.0111179362879\alpha_0^s\alpha_2^s r^3 - 0.60587387125098\alpha_0^s\alpha_4^s r \\
&- 3.1468339532713\left(\alpha_0^s\right)^2\left(\alpha_1^s\right)^2 r \\
&- 4.6296729998974\alpha_0^s\left(\alpha_1^s\right)^2 r^2 - 3.2598096829233\left(\alpha_0^s\right)^2\alpha_2^s r^2 \\
&- 1.2984954850039\alpha_1^s\alpha_3^s r - 1.4700626232957\left(\alpha_0^s\right)^3\alpha_2^s r \\
&+ 4.5507541434397\alpha_0^s\alpha_1^s\alpha_2^s r
\end{aligned}
$$

Absolute order 8.

$$
\begin{aligned}
R_{3,8} = {} & 0.00068532023746048 r^8 + 0.125\alpha_6^s r \\
&+ 4.3619461768873\left(\alpha_0^s\right)^4 r^4 + 0.038536574146577\left(\alpha_0^s\right)^7 r \\
&+ 5.2029254536162\left(\alpha_0^s\right)^5 r^3 + 0.021703788578011\alpha_0^s r^7 \\
&+ 1.545267118531\left(\alpha_0^s\right)^3 r^5 + 1.7512659597646\left(\alpha_0^s\right)^6 r^2 \\
&+ 1.010573413351\left(\alpha_1^s\right)^2 r^4 + 0.26453516473845\left(\alpha_0^s\right)^2 r^6 \\
&+ 0.13887279321988\alpha_4^s r^3 + 0.087120557839057\alpha_2^s r^5 \\
&+ 1.524715047639\left(\alpha_2^s\right)^2 r^2 - 3.8575245524135\left(\alpha_1^s\right)^3 r^2 \\
&- 0.05080989468861\alpha_1^s r^6 - 0.13952053443311\alpha_5^s r^2 \\
&- 0.12024763108045\alpha_3^s r^4 + 1.1971819419038\alpha_0^s\alpha_4^s r^2 \\
&+ 10.320494578314\left(\alpha_0^s\right)^3\alpha_2^s r^2 + 1.4593688359096\alpha_0^s\alpha_2^s r^4 \\
&+ 4.7937562503095\left(\alpha_1^s\right)^2\alpha_2^s r + 10.176804101055\alpha_0^s\left(\alpha_1^s\right)^2 r^3 \\
&+ 3.3523814346221\alpha_0^s\left(\alpha_2^s\right)^2 r + 7.272533332723\left(\alpha_0^s\right)^2\alpha_2^s r^3 \\
&+ 4.3030456716032\left(\alpha_0^s\right)^3\left(\alpha_1^s\right)^2 r + 2.4584809206766\alpha_1^s\alpha_3^s r^2 \\
&+ 1.355409833533\left(\alpha_0^s\right)^2\alpha_4^s r \\
&+ 21.892191226939\left(\alpha_0^s\right)^2\left(\alpha_1^s\right)^2 r^2 \\
&+ 1.5128907607855\left(\alpha_0^s\right)^4\alpha_2^s r - 1.7733901443961\left(\alpha_0^s\right)^3\alpha_3^s r \\
&- 1.0496651482817\alpha_0^s\alpha_1^s r^5 - 4.6393166739312\left(\alpha_0^s\right)^2\alpha_3^s r^2 \\
&- 0.61616427546487\alpha_0^s\alpha_5^s r - 13.841809923869\left(\alpha_0^s\right)^4\alpha_1^s r^2 \\
&- 4.4309930101446\alpha_0^s\left(\alpha_1^s\right)^3 r - 18.743138308883\left(\alpha_0^s\right)^3\alpha_1^s r^3 \\
&- 2.6475317332077\alpha_1^s\alpha_2^s r^3 - 1.3805669230387\alpha_1^s\alpha_4^s r \\
&- 1.5289756803498\alpha_0^s\alpha_3^s r^3 - 0.83496561933704\left(\alpha_0^s\right)^5\alpha_1^s r \\
&- 7.2513766025917\left(\alpha_0^s\right)^2\alpha_1^s r^4 - 1.9358793917918\alpha_2^s\alpha_3^s r \\
&+ 5.4470368262232\alpha_0^s\alpha_1^s\alpha_3^s r - 16.335385878746\alpha_0^s\alpha_1^s\alpha_2^s r^2 \\
&- 9.3326987690427\left(\alpha_0^s\right)^2\alpha_1^s\alpha_2^s r
\end{aligned}
$$